



Fig. 3: Mission-critical electronic equipment

What standards do we have

ESD standards are highly necessary because to objectively evaluate and compare competitive ESD control products. These standards are followed by the manufacturers and users of ESD-sensitive devices and control products, and certification organisations that help them in developing, implementing, auditing and certifying ESD control programmes.

Due to high importance of ESD control in assuring reliability of electronic products, standards for ESD control have evolved pretty quickly. The standards that we have today can be separated in three groups.

The first group consists of those that provide ESD control programme guidance or requirements. These include documents such as *ANSI ESD S20.20-2007* Standard for the Development of an ESD Control Program, *ANSI/ESD S8.1-Symbols-ESD Awareness* or *ESD TR20.20-ESD Handbook*. Such documents give a broad overview of the requirements of ESD control programme. For more specific understanding of each process you will have to go through the second group of standard documents.

The second group covers requirements for specific products or procedures such as packaging and grounding. For example, *ANSI/ESD S6.1* for grounding and *ANSI/ESD S541* for packaging materials for ESD sensitive items.

The third group of documents covers the standardised test methods used to evaluate ESD products and

materials. These include standards such as *ANSI/ESDA-JEDEC JS-001-2010* for device testing, Human Body Model and *ANSI/ESD STM7.1* for resistive characterisation of floor materials to name a few.

Basic ESD control system

Fig. 4 shows a beautiful example of the basic ESD control system where the most crucial elements are indicated. Just by following these five steps a very efficient electrostatic-protected area can be developed to increase the reliability multifold.

1. Personal grounding and isolation. A basic ESD control rule is to ground all conductors including people at the ESD workstation. Grounding works very efficiently in ESD control systems and reliably removes electrostatic charges to ground. For such a grounding system it is important that the electrical wiring system of your lab is correct. Electrical outlets need to be checked to verify the correct wiring of live, neutral and ground. Fig. 4 shows only a wrist strap for grounding the worker, but there are other ways like use of shoe straps.

Also, some clothes are a good conductor of ESD charges. Therefore ESD garments are a good idea for added protection. The basic ESD garments used in the electronics industry are ESD aprons, gloves and shoe-covers.

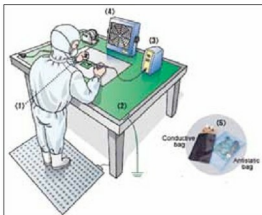


Fig. 4: Basic ESD control system (Image courtesy: HAKKO)

Wrist strap. Standard *ANSI/ESD S1.1-2006* defines wrist strap as an assembled device consisting of a wrist cuff and ground cord that provides electrical connection of a person's skin to ground. The standard document completely describes the parameters for evaluation, acceptance and testing.



Fig. 5: Wrist straps and shoe straps

Though the document describes the whole set of mechanical and electrical parameters over which a wrist strap needs to be evaluated and accepted, the most important parameter among all is the wrist strap continuity and resistance, which should be one megohm ± 20 per cent for acceptance. The document also suggests the testing procedure for the same. While buying these grounding material, do check if they comply with above-mentioned standard and specifications.

ESD garments. The standard document *ANSI/ESD STM2.1* defines test methods for determining the electrical resistance from sleeve to sleeve and point to point of static control garment.

As described in the document, the sleeve-to-sleeve method is intended to test the integrity of the electrical resistance across the seams of the garment, whereas point-